SCALP

Learning Objectives

At the end of lecture student should be able to:

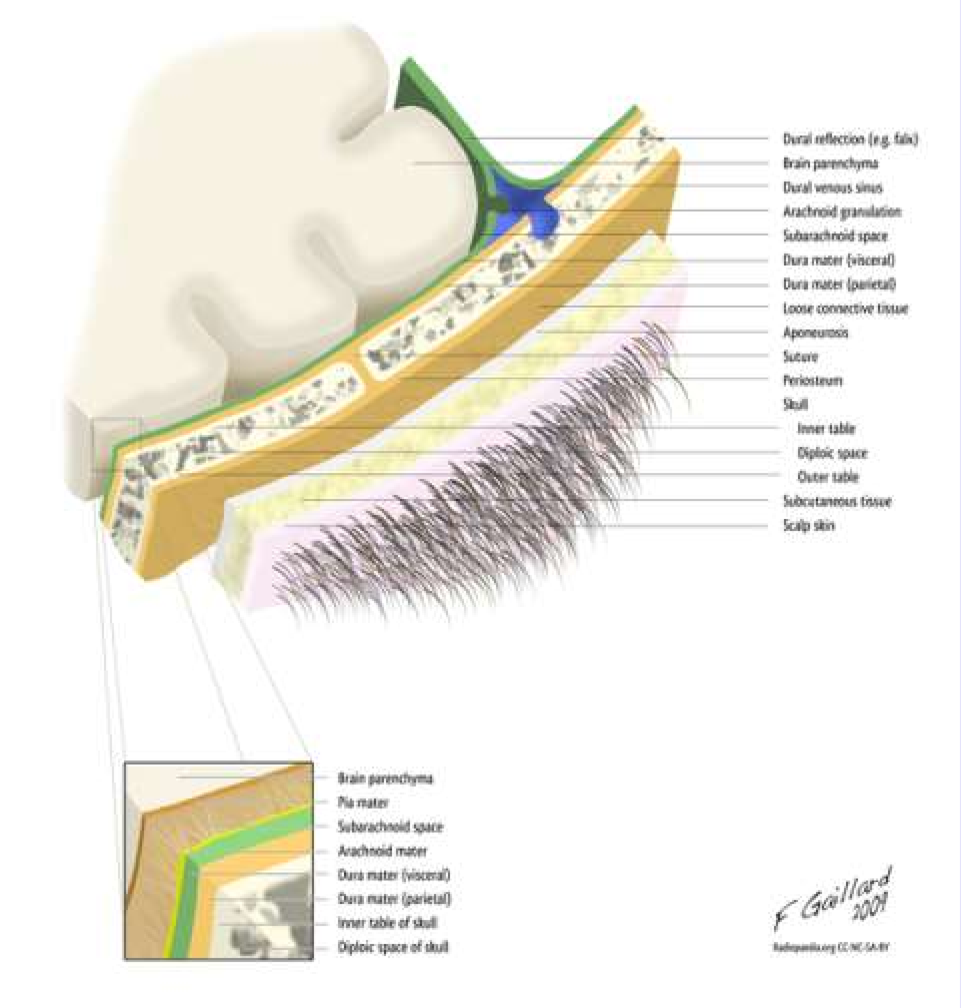
- Know the extent of scalp
- Describe five layers of scalp
- Recall the nerves and vessels of scalp
- Know the clinical correlates

Scalp; Overview

- Soft tissue covering the cranial vault
- It is the hair bearing area of skull
- Anatomically it is the area bordered by the face anteriorly and neck to the sides and posteriorly
- Extend from **supra orbital margin anteriorly to external occipital protuberance & superior nuchal line posteriorly**
- On **each side to superior temporal line**
- Forehead is common to both scalp and face

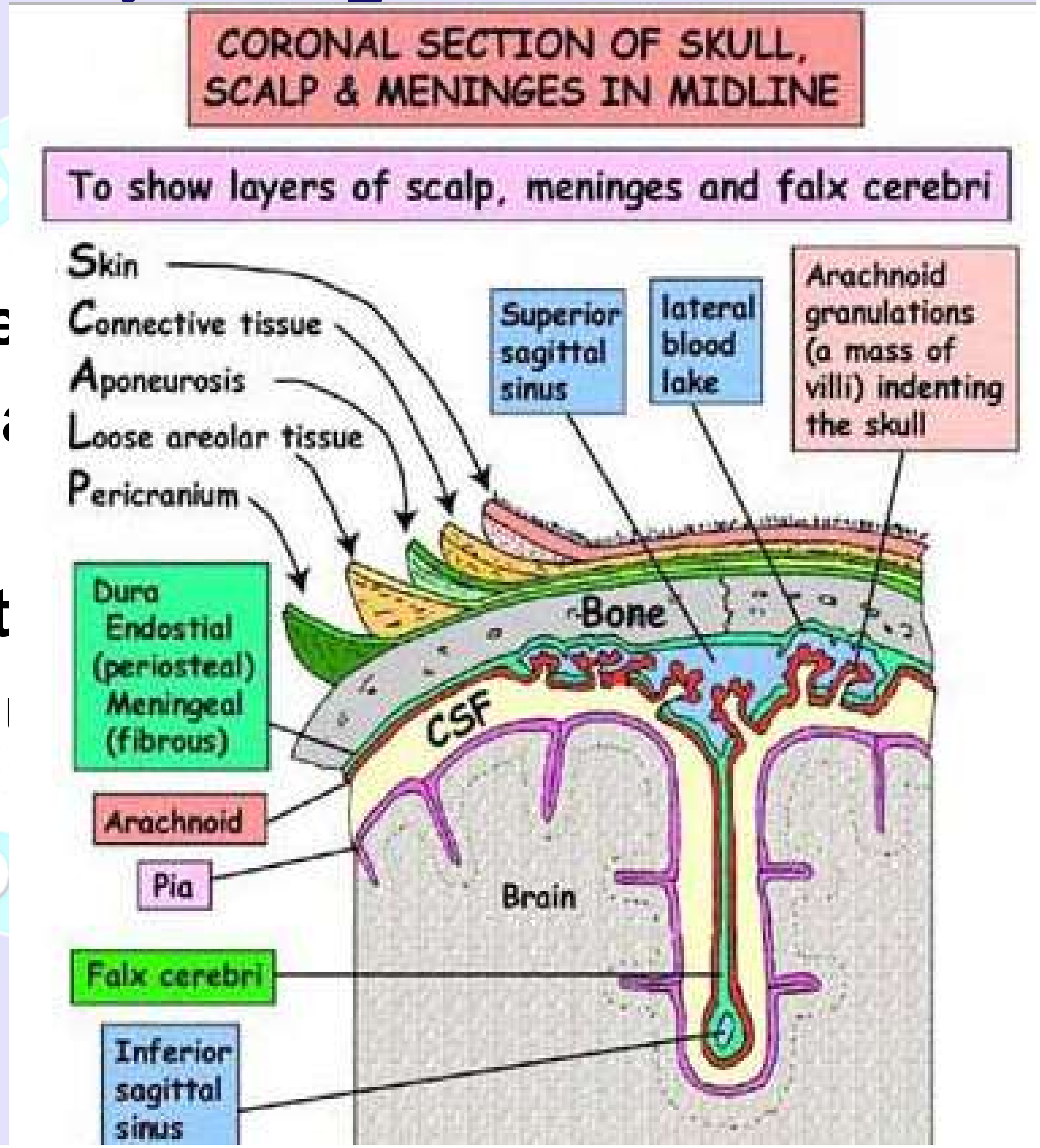
Scalp

- It is usually described as having five layers, which can be remembered with the mnemonic **"SCALP"**



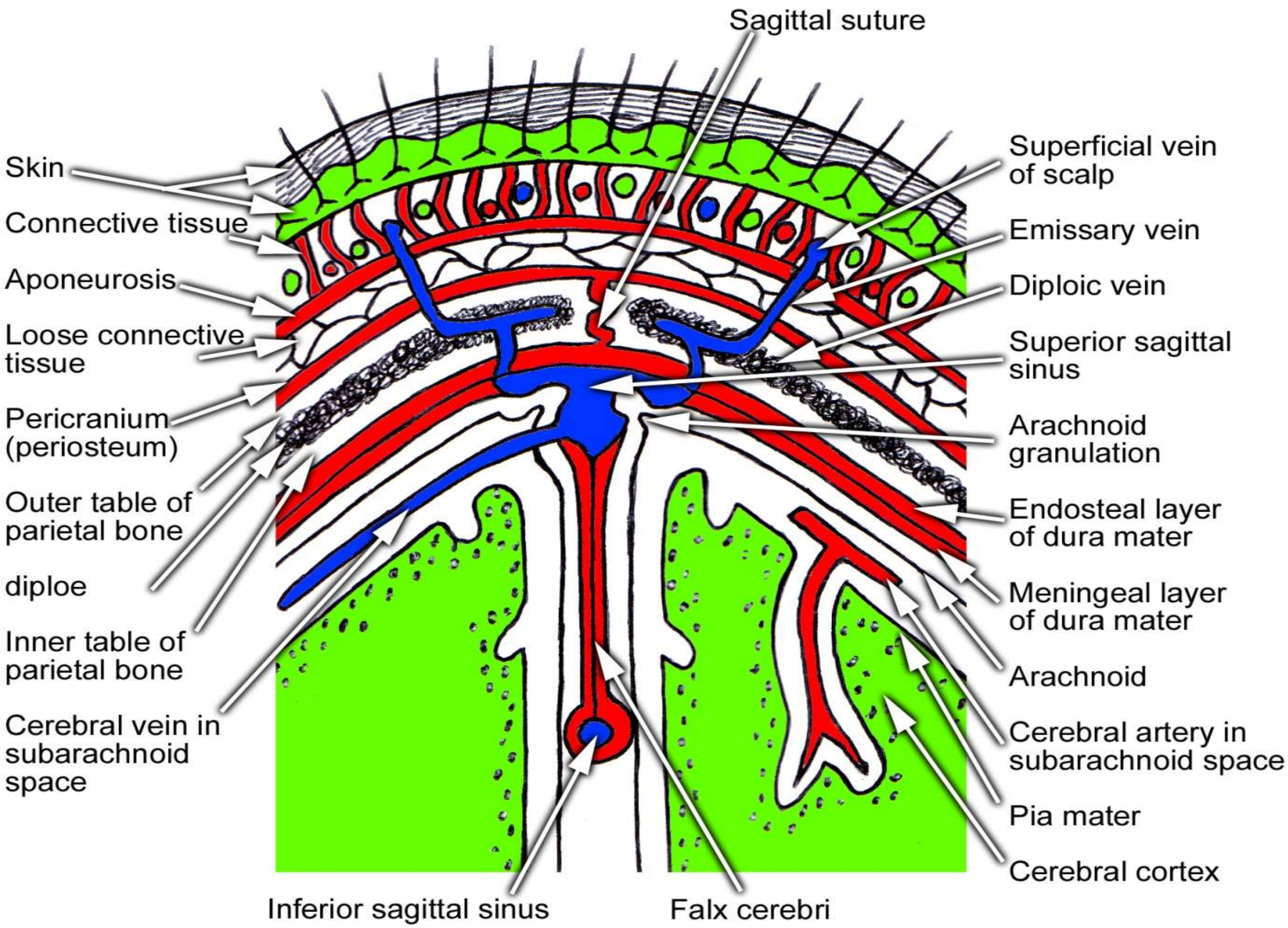
SCALP; Layers

- S-Skin
- C-connective tissue
  (superficial fasci
- A-aponeurosis
  (galea aponeurot
- L-loose areolar tiss
- P-pericranium



Skin

- The skin on the head from which head hair grows.
- Thick and hairy
- Firmly attached to the epicranial aponeurosis through dense fascia
- Abundance of sebaceous glands
- Sebaceous cysts are common

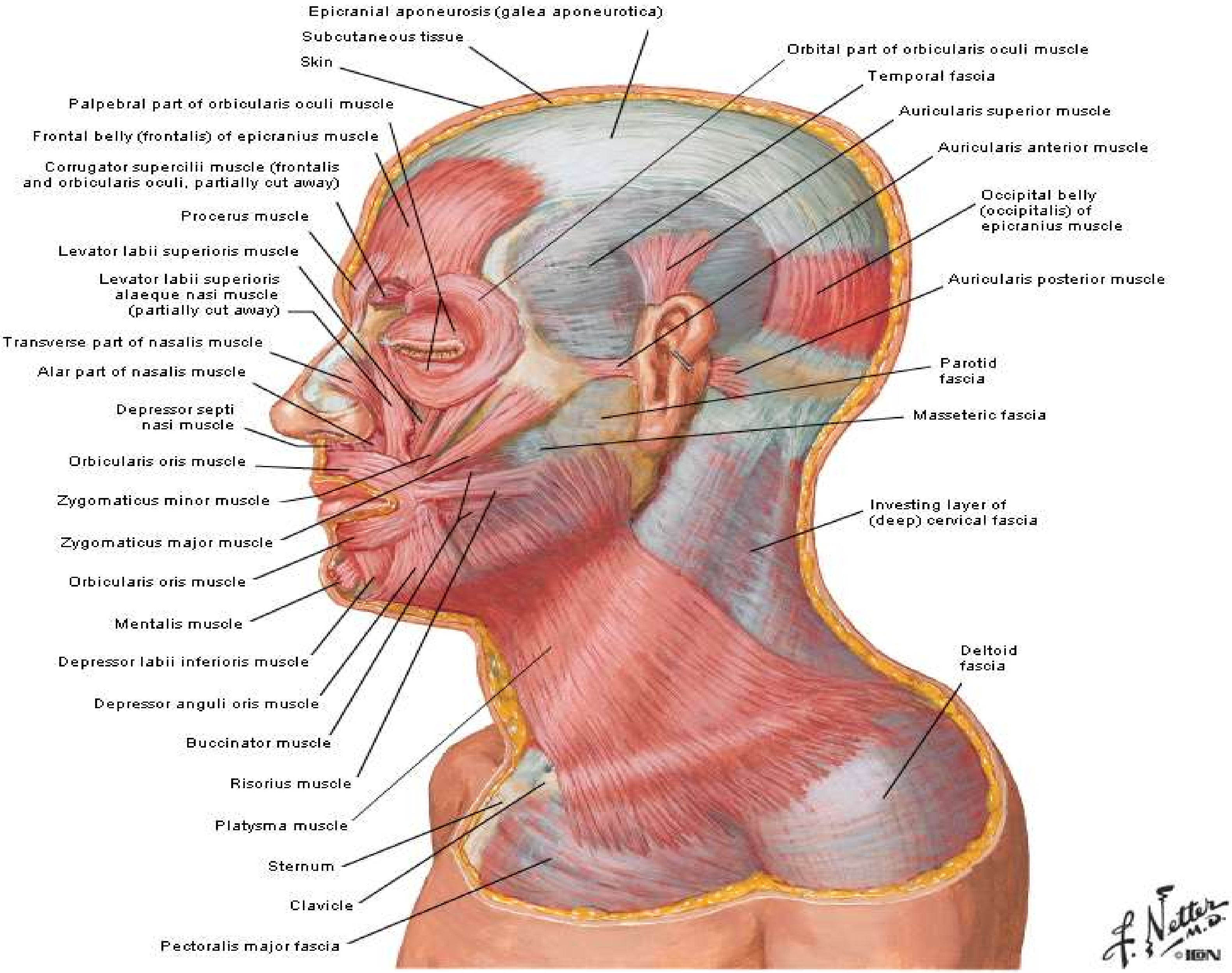

Sagittal suture
Skin
Connective tissue
Aponeurosis
Loose connective tissue
Pericranium (periosteum)
Outer table of parietal bone
diploe
Inner table of parietal bone
Cerebral vein in subarachnoid space
Superficial vein of scalp
Emissary vein
Diploic vein
Superior sagittal sinus
Arachnoid granulation
Endosteal layer of dura mater
Meningeal layer of dura mater
Arachnoid
Cerebral artery in subarachnoid space
Pia mater
Cerebral cortex
Inferior sagittal sinus
Falx cerebri

Connective tissue

- A thin layer of fat and fibrous tissue lies beneath the skin.
- Fibrous and dense containing blood vessels and nerves
- Binds skin to subjacent aponeurosis
- Wounds bleed profusely as blood vessels are prevented from retraction by fibrous tissue. Bleeding is stopped by applying pressure against the bone
- Subcutaneous hemorrhage are not extensive since fascia is dense
- Inflammation cause little swelling but are much painful

Aponeurosis

- The aponeurosis called epicranial aponeurosis (or galea aponeurotica) is the next layer.
- It is a tough layer of dense fibrous tissue which runs from the frontalis muscle anteriorly to the occipitalis posteriorly.
- Frontal belly originate from skin of forehead and mingled with orbicularis oculi muscle
- Occipital belly originate from lateral 2/3 of superior nuchal line
- It gaps if cut transversely and should be stitched



Epicranial aponeurosis (galea aponeurotica)
Subcutaneous tissue
Skin
Palpebral part of orbicularis oculi muscle
Frontal belly (frontalis) of epicranius muscle
Corrugator supercilii muscle (frontalis and orbicularis oculi, partially cut away)
Procerus muscle
Levator labii superioris muscle
Levator labii superioris alaeque nasi muscle (partially cut away)
Transverse part of nasalis muscle
Alar part of nasalis muscle
Depressor septi nasi muscle
Orbicularis oris muscle
Zygomaticus minor muscle
Zygomaticus major muscle
Orbicularis oris muscle
Mentalis muscle
Depressor labii inferioris muscle
Depressor anguli oris muscle
Buccinator muscle
Risorius muscle
Platysma muscle
Sternum
Clavicle
Pectoralis major fascia
Orbital part of orbicularis oculi muscle
Temporal fascia
Auricularis superior muscle
Auricularis anterior muscle
Occipital belly (occipitalis) of epicranius muscle
Auricularis posterior muscle
Parotid fascia
Masseteric fascia
Investing layer of (deep) cervical fascia
Deltoid fascia
F. Netter M.D.
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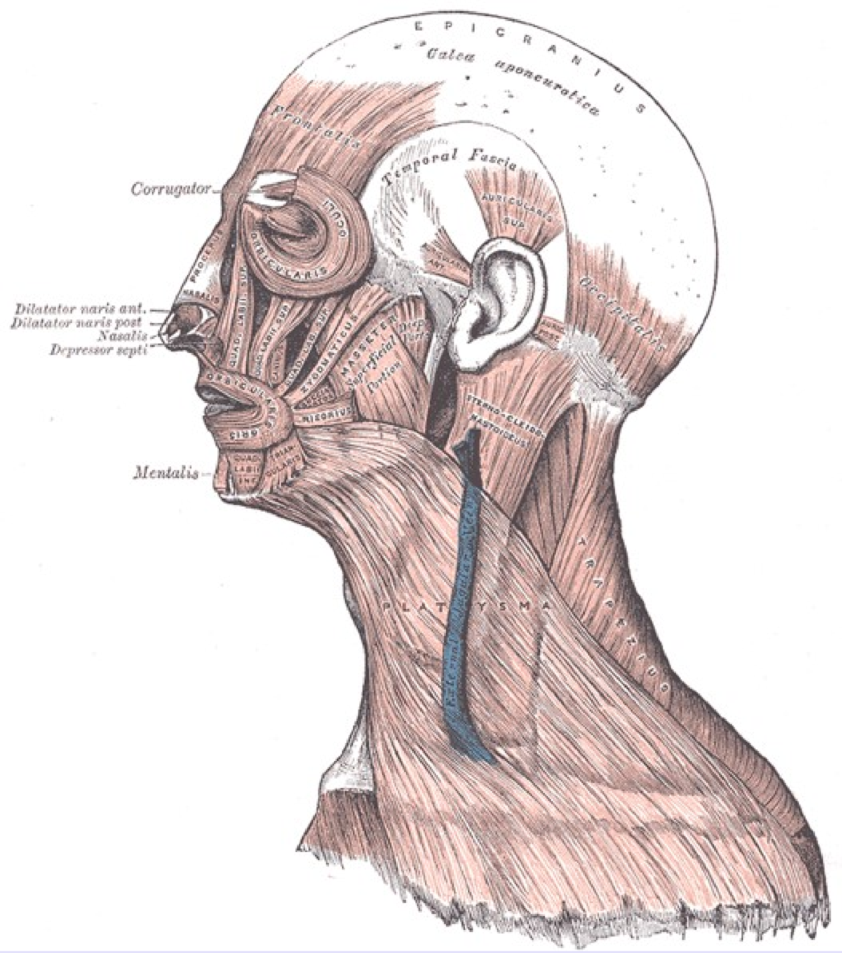
Occipitofrontalis muscle

- **Origin: consists of 2 occipital bellies and 2 frontal bellies. The occipital bellies arise from the superior nuchal lines on the occipital bone. The frontal bellies originate from the skin and superficial fascia of the upper eyelids.**
- **Insertion: all 4 are inserted into the epicranial aponeurosis.**
- **Nerve supply: occipital belly is innervated by the posterior auricular branch of the facial nerve, and frontal belly is innervated by the frontal branch of the facial nerve.**
- **Action: The frontal bellies can raise the eyebrows.**



Loose areolar connective tissue

- provides an easy plane of separation between the upper three layers and the pericranium.
- Extends anteriorly into the eyelids because frontalis has no bony attachment
- Posteriorly to superior nuchal line
- On each side to superior temporal line
- provides a plane of access in craniofacial surgery and neurosurgery.
- Bleeding cause generalized swelling of scalp
- **Called dangerous layer of scalp** because of the ease by which infectious agents can spread through it to emissary veins which then drain into the cranium (venous sinus)
- Bleeding lead **to black eye**

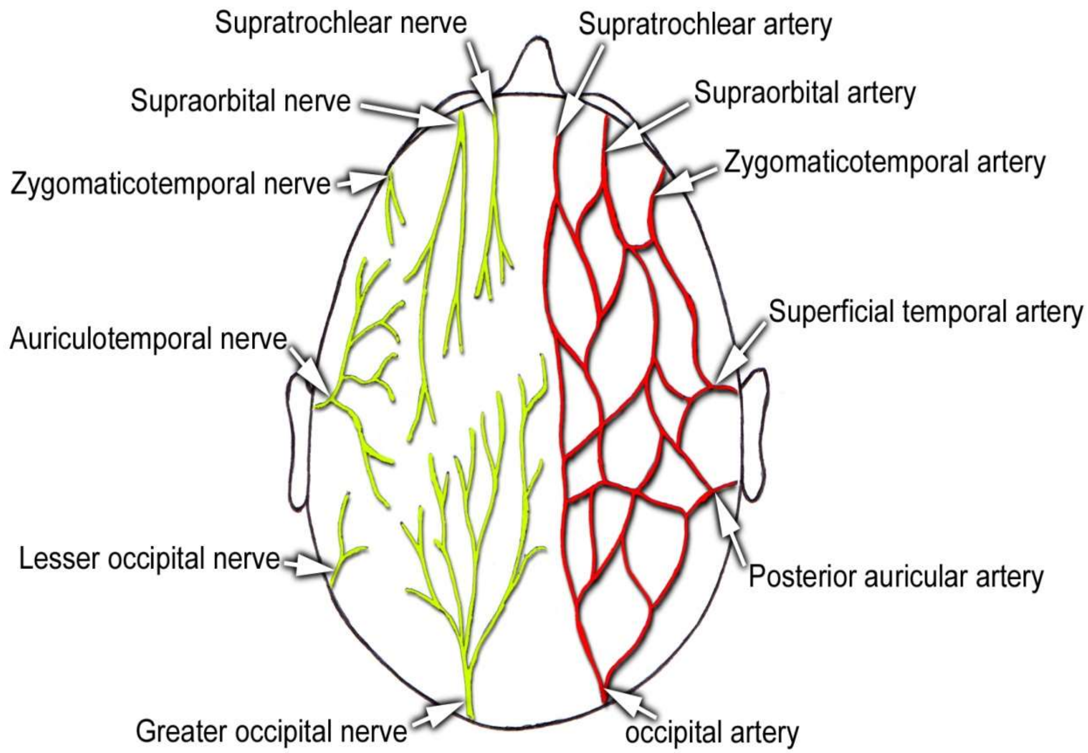
Pericranium

- Is the **periosteum of skull**
- provides nutrition to the bone and the capacity for repair.
- Loosely attached to surface of bone but is firmly adherent to the sutures
- Injury deep to it take the shape of bone **(cephalhaematoma)**

Blood supply Of Scalp

- **Arteries**
  - Supratrochlear
  - Supraorbital
  - Superficial temporal
  - Posterior auricular artery
  - Occipital artery
- **Veins-**

  follows the artery

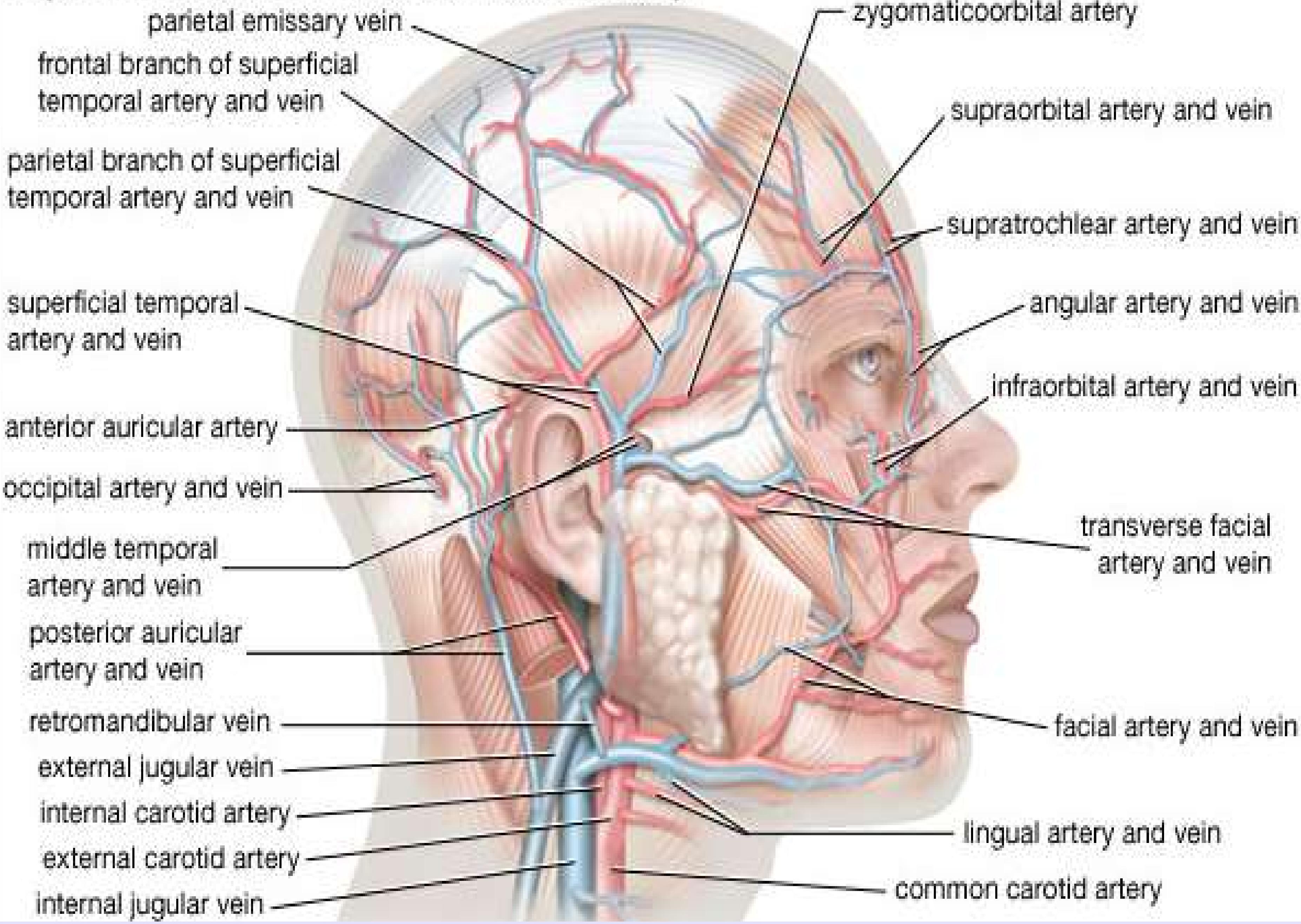


Blood supply Of Scalp

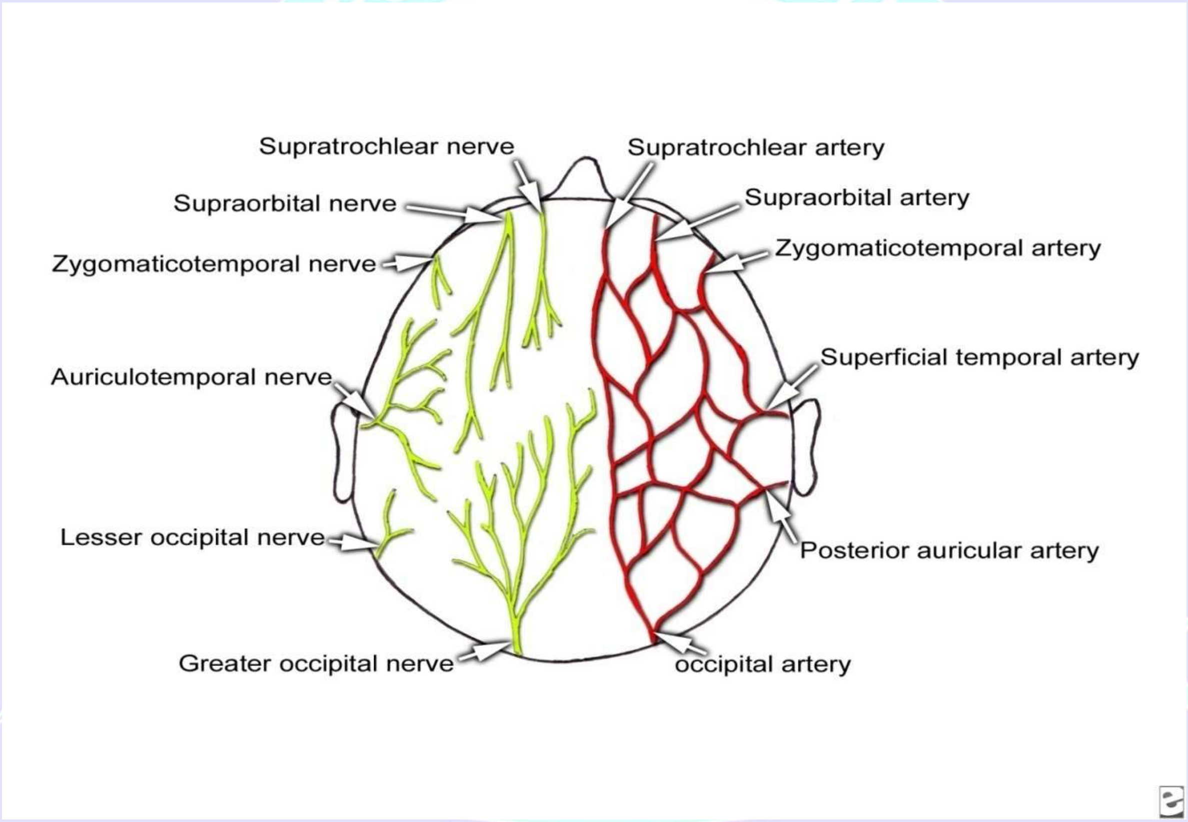
The blood supply of the scalp is via five pairs of arteries, three from external carotid and two from the internal carotid:

- **internal carotid**
  - **the supratrochlear artery is a branch of the ophthalmic artery, to the midline forehead**
  - **the supraorbital artery is a branch of the ophthalmic artery, to the lateral forehead and scalp as far up as the vertex.**
- **external carotid**
  - **the superficial temporal artery gives off frontal and parietal branches to supply much of the scalp**
  - **the occipital artery which runs posteriorly to supply much of the posterior aspect of the scalp**
  - **the posterior auricular artery, a branch of the external carotid artery, ascends behind the auricle to supply the scalp above and behind the auricle.**

Superficial arteries and veins of face and scalp

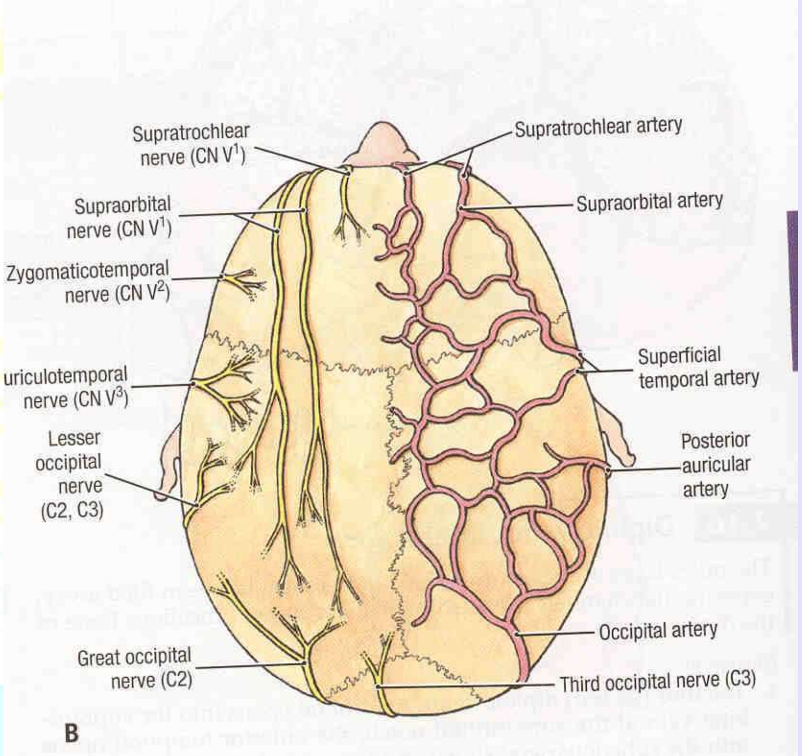


Arteries and nerves of scalp



Nerve supply

- **In front of auricle**
  - Supratrochlear n.
  - Supraorbital n.
  - Zygomaticotemporal n.
  - Auriculotemporal n.



- **Behind auricle**
  - Lesser occipital n.
  - Greater occipital n.
  - Third occipital n.

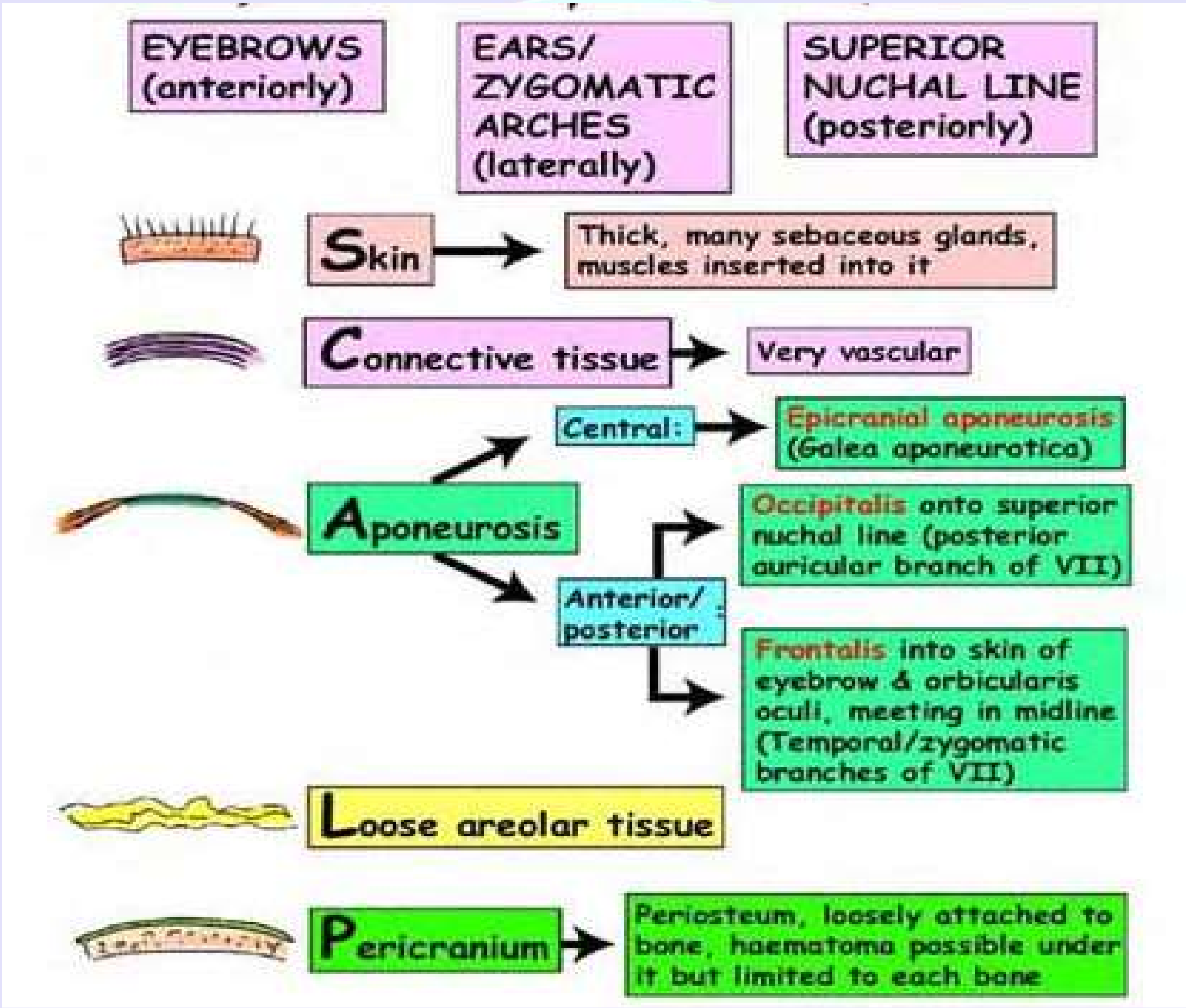
Nerves of scalp

- Supratrochlear nerve and the supraorbital nerve from the ophthalmic division of the trigeminal nerve supply forehead and front of scalp
- Greater occipital nerve (C2) posteriorly up to the vertex
- Third occipital (C3) supply posterior scalp
- Lesser occipital nerve (C2) behind the ear
- Zygomaticotemporal nerve from the maxillary division of the trigeminal nerve supplying the hairless temple
- Auriculotemporal nerve from the mandibular division of the trigeminal nerve also supplies temple

Lymphatics

- **Anterior part**
  - Preauricular (parotid) gr. of lymph node
- **Posterior part**
  - Posterior (mastoid) gr. of lymph node &occipital gr. of lymph node
  - there are no lymph nodes in scalp

Scalp; layers and Extensions



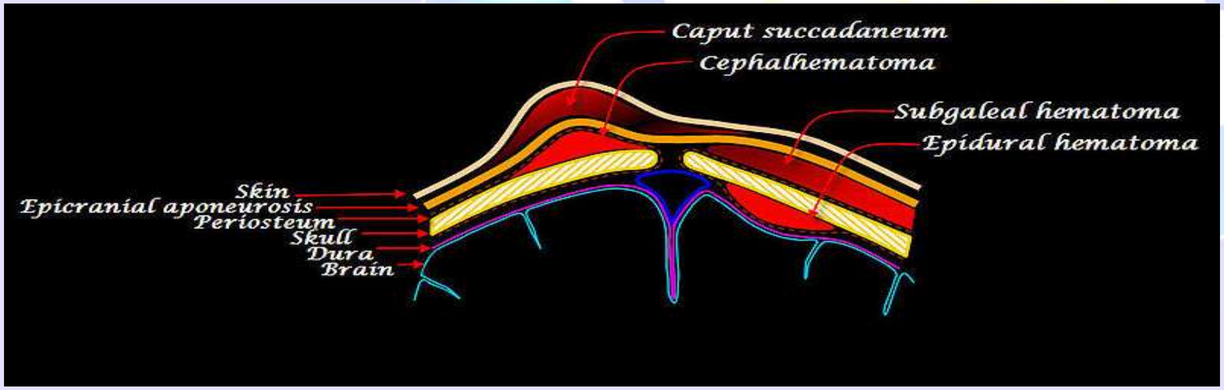
CLINICAL CORELATES

Sebaceous cyst

- **Plentiful sebaceous glands make the scalp one of the most common sites for sebaceous cysts.**
- **Wounds in the scalp bleed profusely because the fibrous fascia prevents vasoconstriction.**
- **The emissary veins do not have valves and open in the loose areolar tissue; therefore, infection can be transmitted from the scalp to the cranial cavity. The layer of**

Cephalhematoma

- It is a hemorrhage of blood between the skull and the periosteum of a newborn baby
- secondary to rupture of blood vessels crossing the periosteum.

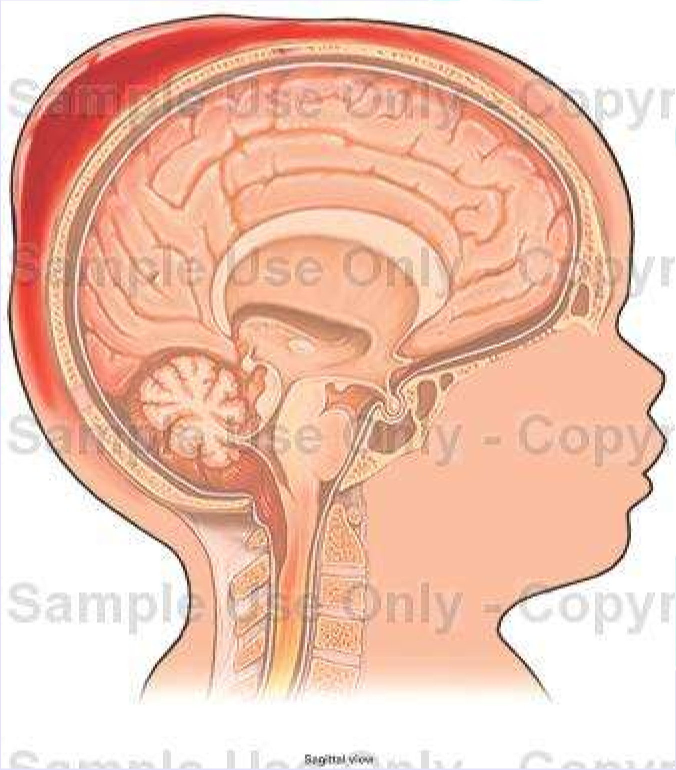


Caput succedaneum

- It is a neonatal condition involving a **subcutaneous, extraperiosteal fluid collection** caused by the pressure of the presenting part of the scalp against the dilating cervix during delivery.
- It involves bleeding below the scalp and above the periosteum.

Comparison Of Caput succedaneum and Cephalhematoma

- Caput succedaneum



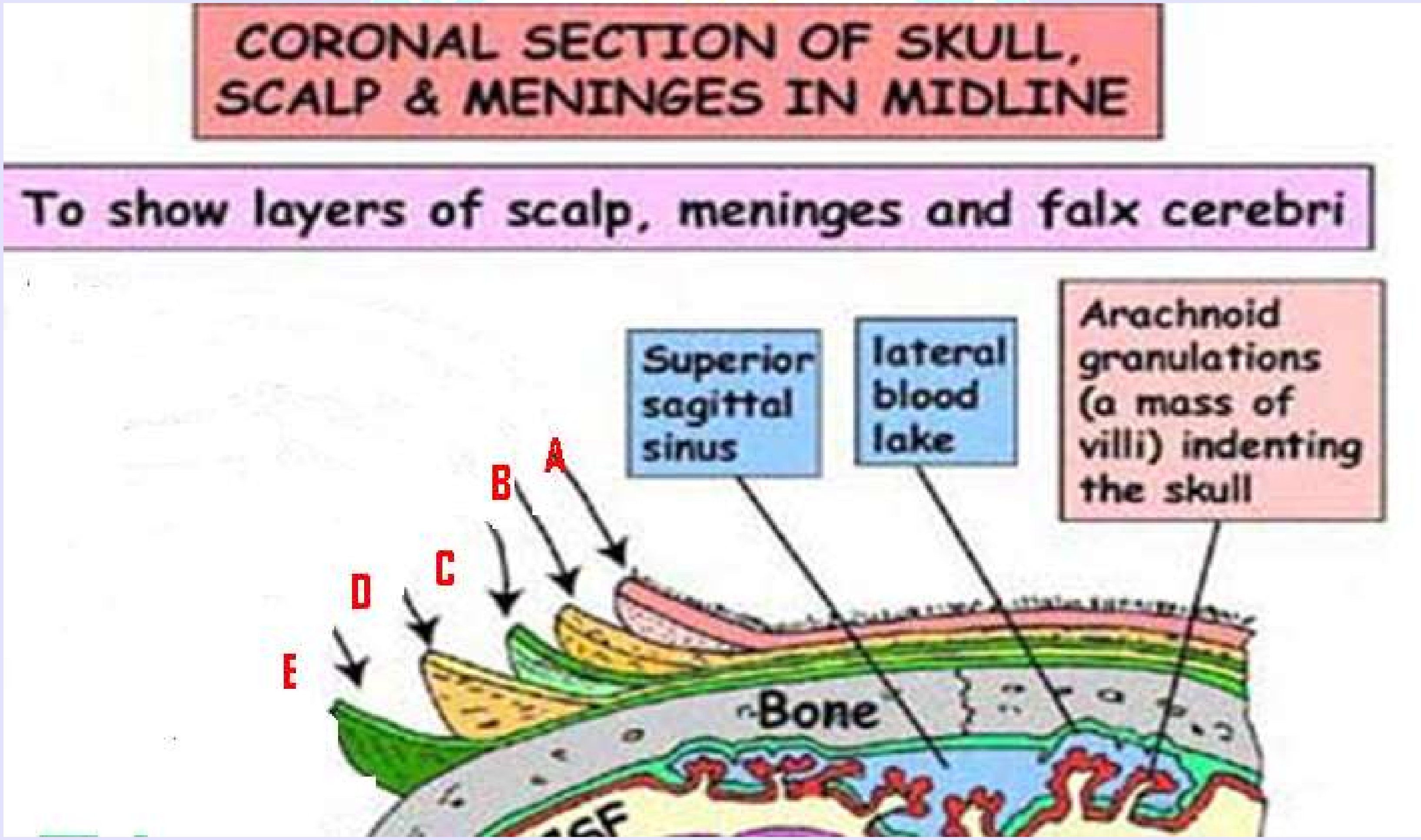
- cephalhaematoma

Psoriasis
Black eye

Self Assessment

Q1. What is scalp? Give the layers of Scalp.

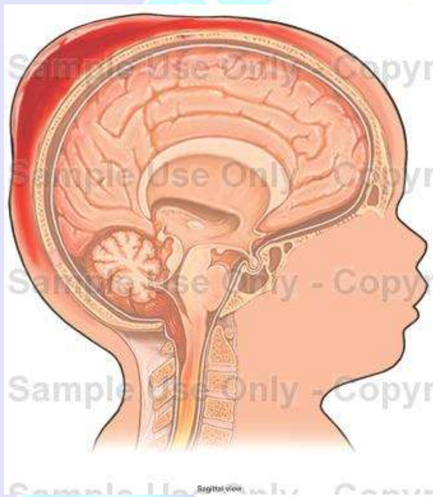
Q2. Identify the layers marked with arrows A,B,C,D and E in the figure.



Self Assessment

Q3. What is the arterial supply of scalp?

Q4.Identify the following clinical conditions?



Q5.What is the difference b/w cephalhematoma and Caput succedaneum?

Thanks